Translation

SAT Math · Foundations to Mastery · Week of 18 October

ANSWER KEY — tutor copy

How to use this sheet

This is four short sittings, not one long one. Four half-hours across the week will do more for you than one long Saturday, and you are far more likely to finish it.

- **Part A is no calculator.** None. It is 15 minutes and it keeps the last three weeks alive.
- **Part B asks you not to solve.** You write the equation and stop. The arithmetic is not the part costing you marks.
- When you get one wrong, write it in your error log: what you did, and what you should have done.

Sitting	Part	What it drills	Minutes
Sunday	A	Retention: signs, slope, systems	15
Tuesday	B	Setups only — do not solve	25
Thursday	C	The two traps	25
Saturday	D, E	Set up and solve; six mixed, timed	27

Part A — Retention

No calculator. 15 minutes.

SUNDAY

Three weeks of work sits behind these ten. Keep the brackets around every substitution.

1. $-9-(-4)=$ -5
2. Expand: $-3(x-5)$ $=-3x+15$
3. Expand and simplify: $-(4x-7)+x$ $=-3x+7$
4. A line passes through $(-2,7)$ and $(2,-1)$. What is its slope? -2. $\dfrac{-1-7}{2-(-2)}=\dfrac{-8}{4}=-2$.
5. A line is given by $3x+4y=8$. What is its slope? $-3/4$. $y=-\dfrac{3}{4}x+2$.
6. A line is given by $5x-2y=10$. What is the y-coordinate of its y-intercept? -5. **Put** $x=0$**:** $-2y=10$.
7. Solve by substitution: $y=2x-3$ and $3x+y=12$. Give x and y.

   Answer: _____ *(write your answer)*

   $x=3$, $y=3$. $3x+(2x-3)=12$, **so** $5x=15$. **Check:** $3(3)+3=12$.
8. Solve by elimination: $4x+y=14$ and $2x-y=4$. Give x and y.

   Answer: _____ *(write your answer)*

   $x=3$, $y=2$. **Adding the two equations kills** y**:** $6x=18$. **Then** $y=14-12=2$.
9. Solve $2x+7<1$.

   Answer: _____ *(write your answer)*

   $x<-3$. $2x<-6$. **No flip here — you divided by a positive. Check:** $x=-4$ **gives** $-1<1$**, true.**

10. Solve $5 - 3x \geq 20$.

    Answer: _____ *(write your answer)*

    $x \leq -5$. $-3x \geq 15$, **then divide by** -3 ***and flip*****. Check:** $x = -6$ **gives** $23 \geq 20$**, true.**

Marking rule: below 8 of 10 means an earlier week has decayed. Find which one — items 1–3 are Week 1, items 4–6 are Week 2, items 7–10 are Week 3 — and repair that one before any translation work.

Part B — Setups only

Calculator allowed. 25 minutes.

TUESDAY

Do not solve any of these. For each one you write two things and then stop: first name the unknown on the *Let* line, then write the equation (or the pair of equations) on the line under it. No arithmetic. No answer.

You can do the arithmetic already. What costs you marks is building the equation, so this sitting practises only that. An equation you reached without naming the unknown does not count this week.

1. A moving company charges a flat \$75 for the van plus \$0.60 for every mile driven. A move cost \$153 in total. Write an equation for the number of miles.
   Let m **be the number of miles.** $75 + 0.60m = 153$**. (True** $m = 130$**:** $75 + 0.60(130) = 153$**.)**

2. A printing shop charges an \$18 setup fee plus \$0.35 for each flyer. An order cost \$81. Write an equation for the number of flyers.
   Let f **be the number of flyers.** $18 + 0.35f = 81$**. (True** $f = 180$**:** $18 + 0.35(180) = 81$**.)**

3. A theatre sold 40 tickets and took \$395. Adult tickets cost \$12 and child tickets cost \$7. Write a *pair* of equations.
   Let a **be the number of adult tickets and** c **the number of child tickets.** $a + c = 40$ **and** $12a + 7c = 395$**. (True** $a = 23$**,** $c = 17$**:** $23 + 17 = 40$ **and** $276 + 119 = 395$**.)**

4. A box holds 30 pens. Black pens cost \$1.20 each and blue pens cost \$0.80 each, and the whole box is worth \$30. Write a *pair* of equations.
   Let b **be the number of black pens and** u **the number of blue pens.** $b + u = 30$ **and** $1.20b + 0.80u = 30$**. (True** $b = 15$**,** $u = 15$**:** $18 + 12 = 30$**.)**

5. A train travels at a steady 56 miles per hour and covers 280 miles. Write an equation for the time taken.

   Let h **be the number of hours.** $56h = 280$**. (True** $h = 5$**:** $56(5) = 280$**.)**

6. A tank holds 540 litres and a pump removes 18 litres each minute. At some moment 90 litres remain. Write an equation for the time the pump has run.
   Let m **be the number of minutes.** $540 - 18m = 90$**. (True** $m = 25$**:** $540 - 450 = 90$**.) The pump *removes*, so the term is subtracted.**

7. In one class, 35% of the students chose Spanish, and 14 students chose Spanish. Write an equation for the number of students in the class.
   Let n **be the number of students.** $0.35n = 14$**. (True** $n = 40$**:** $0.35(40) = 14$**.)**

8. **Reminder: you are still not solving these.** A jacket is reduced by 20% and sells for \$68. Write an equation for the original price.
   Let p **be the original price in dollars.** $0.80p = 68$**. (True** $p = 85$**:** $0.80(85) = 68$**.) A 20% reduction leaves 80%, so the multiplier is** 0.80**, not** 0.20**.**

9. A club raised its membership fee by 12%, and the new fee is \$504. Write an equation for the old fee.
   Let f **be the old fee in dollars.** $1.12f = 504$**. (True** $f = 450$**:** $1.12(450) = 504$**.) An increase of 12% multiplies by** 1.12**.**

10. Three consecutive integers add to 84. Write an equation.
Let n be the smallest integer. $n+(n+1)+(n+2)=84$. **(True $n=27$: $27+28+29=84$.)**

11. Two consecutive odd integers add to 92. Write an equation.
Let n be the smaller odd integer. $n+(n+2)=92$. **(True $n=45$: $45+47=92$.) Consecutive *odd* integers differ by 2, not 1.**

12. A rectangle has a length that is 5 cm more than twice its width, and its perimeter is 70 cm. Write an equation in the width alone.
Let w be the width in cm, so the length is $2w+5$. $2(2w+5)+2w=70$. **(True $w=10$, length 25:** $2(25)+2(10)=70$.**)**

Part C — The two traps

Calculator allowed. 25 minutes.

THURSDAY

Two mistakes account for most of what goes wrong when you translate.

The reversal trap. "7 less than x" is $x-7$. Never $7-x$. The thing you start with is written first, and the amount taken away comes after the minus sign.

The direction trap. "Twice as many apples as pears", with a apples and p pears, is $a=2p$. The multiplier goes on the *smaller* side, because that is the side that needs building up to match.

1. Which expression represents "seven less than a number x"?
A) $7-x$ B) $x-7$ C) $7x$ D) $\frac{7}{x}$
B. A is the reversal: it reads the words in the order they appear. C reads "less" as a product. D reads it as a quotient.

2. A number n decreased by 12 is equal to 5 times the number. Which equation says that?
A) $12-n=5n$ B) $n-12=5n$ C) $n-12=n+5$ D) $12-n=\frac{n}{5}$
B. "Decreased by" takes 12 off n, so n comes first. A reverses the subtraction and gives $n=2$ instead of $n=-3$. C reads "5 times" as "plus 5". D does both wrongs at once.

3. Which expression represents "9 subtracted from 4 times k"?
A) $9-4k$ B) $4k-9$ C) $4(k-9)$ D) $4-9k$
B. "Subtracted from" means the 9 comes off the $4k$. A is the reversal. C subtracts before multiplying. D swaps the two numbers.

4. A number a is 6 less than three times a number b. Which equation says that?
A) $a=6-3b$ B) $a=3b-6$ C) $a=3(b-6)$ D) $6-a=3b$
B. Build $3b$ first, then take 6 off it. A is the reversal. C takes the 6 off before multiplying. D takes the 6 off the wrong quantity.

5. A student is told "the cost c is \$8 less than the price p" and writes $c=8-p$. In one sentence, say why that is wrong, then write the correct equation.

It is the reversal trap: the 8 is what gets taken off the price, so the price must come first. Correct: $c=p-8$. Test it — a price of \$20 gives a cost of \$12, while $8-20$ gives $-\$12$, and a cost cannot be negative.

6. A shelter has three times as many cats c as dogs d. Which equation says that?
   A) $c = 3d$ B) $3c = d$ C) $c + 3 = d$ D) $c = d + 3$
   A. Test with 4 dogs: there are 12 cats, and $12 = 3(4)$ holds. B is the direction trap — it would put 12 dogs with 4 cats. C and D read "times" as "more than".

7. A theatre has 5 times as many standard seats s as premium seats p. Which equation says that?
   A) $5s = p$ B) $s = 5p$ C) $s = p + 5$ D) $p = s + 5$
   B. Test with 30 premium seats: 150 standard, and $150 = 5(30)$ holds. A is the direction trap. C and D turn a multiplier into an addition.

8. A recipe uses twice as much flour f as sugar g, both measured in cups. Which equation says that?
   A) $f = 2g$ B) $2f = g$ C) $f = g + 2$ D) $fg = 2$
   A. Test with 3 cups of sugar: 6 cups of flour, and $6 = 2(3)$ holds. B is the direction trap. C reads "twice" as "2 more". D multiplies the two amounts together.

9. A class has 4 times as many juniors j as seniors n. A student writes $4j = n$. In one sentence, say why that is wrong, then write the correct equation.

   _____

   _____

   It points the multiplier the wrong way: $4j = n$ says the seniors are the larger group. Correct: $j = 4n$. Test it — 5 seniors give 20 juniors, and $20 = 4(5)$ holds, while $4(20) = 80$ is nowhere near 5.

10. A jar has half as many red marbles r as blue marbles b. A student writes $r = 2b$. In one sentence, say why that is wrong, then write the correct equation.

    _____

    _____

    It doubles the group that is meant to be the smaller one. Correct: $r = \frac{b}{2}$, which can also be written $b = 2r$. Test it — 10 blue marbles give 5 red, while $r = 2b$ would give 20.

Part D — Set up and solve

Calculator allowed. 15 minutes.

SATURDAY

Now you do both halves. Name the unknown, write the equation, solve it, and then read your answer back against the story to check it makes sense.

1. A gym charges a one-time joining fee of \$40 plus \$29 for each month of membership. A member has paid \$359 in total. For how many months has he been a member?
   Answer: _____ *(write your answer)*
   11. Let m be the months: $40 + 29m = 359$, so $29m = 319$ and $m = 11$. Check: $40 + 29(11) = 359$.

2. A theatre sold 60 tickets and took \$732. Adult tickets cost \$15 and student tickets cost \$9. How many student tickets were sold?
   Answer: _____ *(write your answer)*
   28. Let a and s be the counts: $a + s = 60$ and $15a + 9s = 732$. Substituting $a = 60 - s$ gives $900 - 6s = 732$, so $s = 28$ and $a = 32$. Check: $480 + 252 = 732$.

3. Three consecutive integers add to 132. What is the greatest of the three?

Answer: _____ *(write your answer)*

45. **Let n be the smallest: $3n+3=132$, so $n=43$ and the three are 43, 44, 45. The question asks for the greatest, not for n.**

4. In a club, 45% of the members voted in an election, and 27 members voted. How many members does the club have?

   Answer: _____ *(write your answer)*

   60. **Let n be the membership: $0.45n=27$, so $n=60$. Check: $0.45(60)=27$.**

5. A rectangle has a length that is 3 cm less than twice its width, and its perimeter is 54 cm. What is its width, in cm?

   Answer: _____ *(write your answer)*

   10. **Let w be the width, so the length is $2w-3$: $2(2w-3)+2w=54$ gives $6w-6=54$ and $w=10$. The length is 17. Check: $2(17)+2(10)=54$.**

6. A tank holding 400 litres drains at 7 litres per minute. After how many minutes do 155 litres remain?

   Answer: _____ *(write your answer)*

   35. **Let m be the minutes: $400-7m=155$, so $7m=245$ and $m=35$. Check: $400-245=155$. Reporting 245 is the trap — that is the water lost, not the time.**

Part E — Mixed, timed

Calculator allowed. 12 minutes, timed.

SATURDAY

Set a timer for 12 minutes. Answer every one — there is no penalty for a wrong answer, so a blank is a mark thrown away. Then check each answer by reading it back against the story.

1. A repair shop charges a flat \$25 inspection fee plus \$0.90 for each minute of labour. Which equation gives the total cost C, in dollars, of a repair taking t minutes?

   A) $C=25t+0.90$ B) $C=0.90t+25$ C) $C=25.90t$ D) $C=0.90(t+25)$

   B. The per-minute charge multiplies t; the fee is charged once. A swaps the two numbers. C charges the fee every minute. D does the same inside a bracket.

2. A shop sells pens at \$2 each and markers at \$5 each. Tom bought 12 items and paid \$39. How many markers did he buy?

   Answer: _____ *(write your answer)*

   5. **Let p and m be the counts: $p+m=12$ and $2p+5m=39$. Substituting gives $24+3m=39$, so $m=5$ and $p=7$. Check: $14+25=39$.**

3. A camp has 4 times as many campers c as counsellors n. Which equation describes this?

   A) $4c=n$ B) $c=4n$ C) $c=n+4$ D) $n=c+4$

   B. Test with 6 counsellors: 24 campers, and $24=4(6)$ holds. A is the direction trap. C and D read the multiplier as an addition.

4. Twelve less than 5 times a number is 63. What is the number?

   Answer: _____ *(write your answer)*

   15. **$5x-12=63$, so $5x=75$ and $x=15$. Check: $75-12=63$. Writing $12-5x=63$ is the reversal trap and gives -10.2.**

5. A rectangle has width w and a length that is 4 more than three times the width. Which expression gives its perimeter?

A) $8w+8$ B) $4w+4$ C) $6w+4$ D) $3w+4$

A. The length is $3w+4$, so the perimeter is $2w+2(3w+4)=8w+8$. Check with $w=5$: sides 5 and 19 give 48, and $8(5)+8=48$. C doubles only the width term. D is the length alone. B adds the two dimensions without doubling either.

6. A jacket is marked down by 30% and sells for \$84. What was its original price, in dollars?

Answer: _____ *(write your answer)*

120. A 30% markdown leaves 70%, so $0.70p=84$ and $p=120$. Check: $0.70(120)=84$. Taking 30% of 84 and adding it gives \$109.20, which does not check.

Five-minute marking routine *Tutor*

1. **Score Part A only.** Below 8/10 means an earlier week has decayed, and the next warm-up repairs that week before anything new is introduced.
2. **In Part B check two things: whether he named the unknown, and whether the equation reads back as the story.** A right equation reached without naming the unknown is still a miss this week. Read his equation aloud as a sentence and ask him whether that is what the question said.
3. **Pick two wrong answers from Part E** and have him narrate his thinking out loud. Leave the rest unmarked.

If the error log is filled in, say so out loud. He will do more of what you visibly value.